SANTHOSH BELL

Junior Robotics Engineer | 3D Concrete Printing Specialist

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PROFESSIONAL SUMMARY

Robotics and Automation Engineering graduate with 1 year of hands-on industrial experience specializing in ABB IRB 6700 industrial robot operations and Robotic 3D Concrete Printing (3DCP) applications. Proven track record in operating ABB IRC5 controllers, utilizing FlexPendants for manual jogging, executing RAPID programs, and assisting with precise TCP and WorkObject calibrations. Highly skilled in tracking material flow, optimizing concrete rheology, and managing on-site production workflows. Seeking immediate opportunities as a 3D Construction Printer Operator or Robotic Automation Technician in the UAE to support the Dubai 3D Printing Strategy.

CORE COMPETENCIES

- **Industrial Robotics:** ABB IRB 6700 Operation, ABB IRC5 Controller, FlexPendant Jogging (Joint/Linear/Reorient)
- **Additive Manufacturing:** 3D Concrete Printing (3DCP), Material Flow Control, Nozzle Troubleshooting
- **Calibration & Setup:** Tool Center Point (TCP) & WorkObject Calibration Assistance, Robot Shifting & Site Preparation
- **Software & Programming:** RAPID Program Execution, Python, C++, ROS, Gazebo Simulation, Slicing Logic
- **Embedded Systems:** Arduino, ESP32, Raspberry Pi 5, Sensor Interfacing, Communication Protocols (UART, I2C, SPI)
- **Documentation & Safety:** Daily Progress Reports (DPR), Industrial Safety Procedures, Preventive Maintenance

PROFESSIONAL EXPERIENCE

Junior Robotics Engineer | Simpliforge 3D Creation Private Limited **August 2026 – Present**

Industry: Construction Robotics / 3D Concrete Printing | Robot: ABB IRB 6700 | Controller: ABB IRC5

- Operated high-payload ABB IRB 6700 robotic arms via the ABB FlexPendant for large-scale, automated 3D concrete printing projects.
- Executed and monitored existing RAPID programs, managing program pointers and system overrides mid-print to ensure zero downtime.
- Assisted senior engineers in precise TCP calculation and WorkObject setup, accommodating varied nozzle geometries and print-bed layouts.
- Supervised layer-by-layer mortar extrusion, monitored concrete mix consistency/rheology, and coordinated with the pumping unit to prevent nozzle blockages.
- Managed post-print maintenance, including thorough system flushing, nozzle cleaning, and comprehensive diagnostic logging of the IRC5 controller.
- Maintained strict adherence to industrial safety standards on-site and filed detailed Daily Progress Reports (DPR).

Robotics & Artificial Intelligence Intern | Logiprompt Pro Academy **September 2025 – May 2026**

- Developed embedded system applications using Arduino and Raspberry Pi for automated smart systems.
- Integrated sensors and motor drivers to build AI-driven automation workflows.

TECHNICAL PROJECTS

- **AI Voice Assistant Robot (Raspberry Pi 5):** Designed an interactive voice-controlled robot featuring real-time motor control, an OLED display interface, and microphone-speaker integration.
- **Smart Home Automation System (ESP32 & IoT):** Implemented an energy-efficient automation network utilizing MQTT and HTTP protocols for remote appliance management and sensor tracking.
- **Autonomous Obstacle Avoidance Robot (Arduino):** Engineered a navigation system using ultrasonic sensor grids to process obstacle parameters and execute real-time routing decisions.

EDUCATION

- **B.E. in Robotics and Automation Engineering** | Loyola Institute of Technology and Science 2021 – 2025
- **Higher Secondary Certificate (HSC)** | Walker Higher Secondary School 2020 – 2021

CERTIFICATIONS

- Robotics with Artificial Intelligence Internship Certification – Logiprompt Pro Academy
- Networking Training Program – AB Technologies, Nagercoil (2023)
- Advanced Smartphone Services & Troubleshooting – New Technology Institute, Coimbatore (2023)

LANGUAGES

Tamil (Native) | **English** (Professional Working Proficiency)